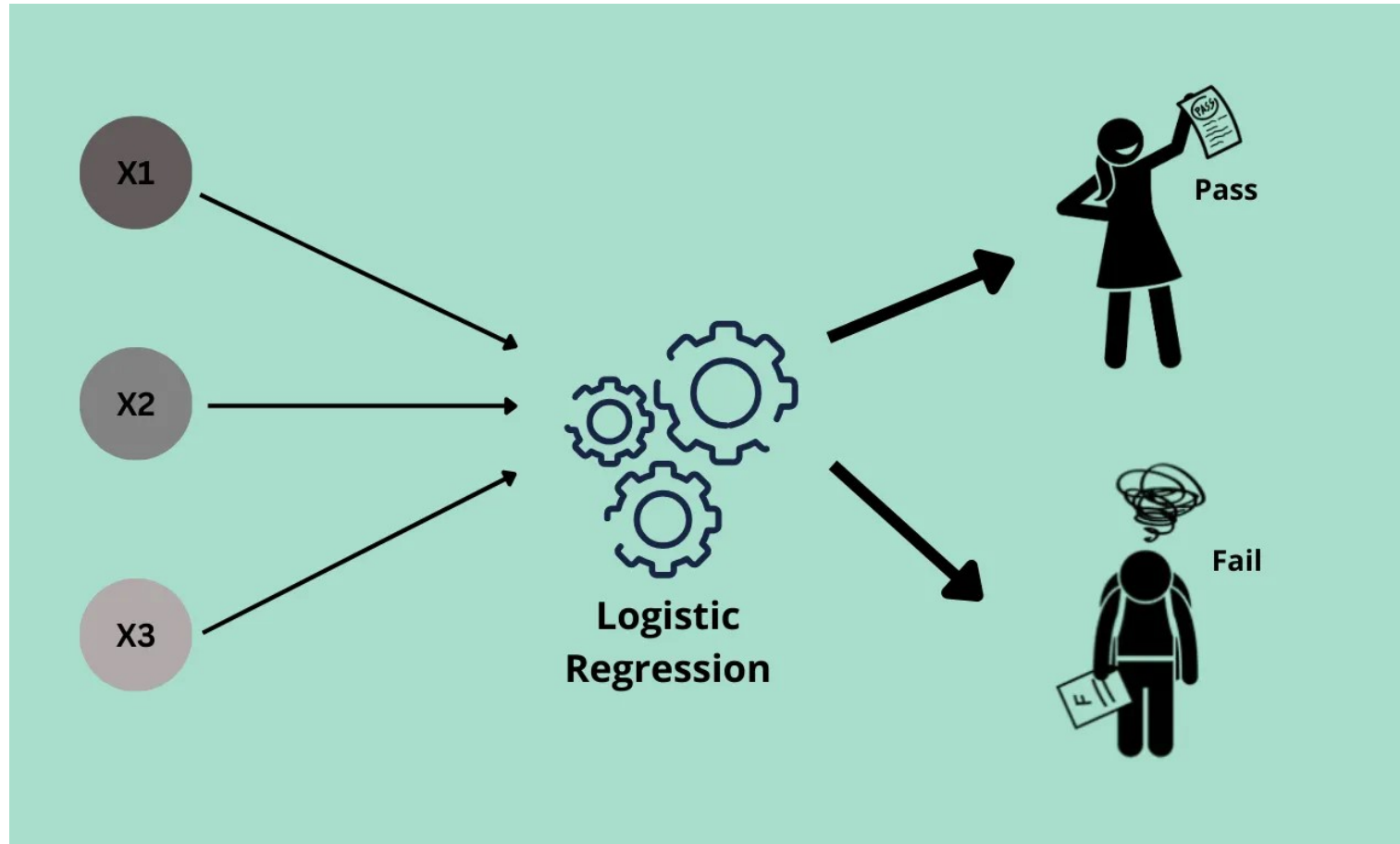


Logistic Regression

Logistic Regression

- Logistic regression is a statistical method used for binary classification.
- Logistic Regression allows us to compute a number that we can interpret as the instance's probability of being part of a class.
- It predicts the probability that an instance belongs to one of two classes by modeling the relationship between independent variables (features) and a binary dependent variable (output).

Binary Logistic Regression



Types of Logistic Regression

- Binary logistic regression or binomial logistic regression: Response variable has two outcomes (e.g., Yes/No, Spam/Not Spam, Pass/Fail, Benign/Malignant, Approved/Rejected)
- Multinomial Logistic Regression: When the response variable has three or more outcomes (e.g., predicting weather: sunny, rainy, or snowy).
- Ordinal logistic regression can be binary or multinomial outcomes but in order like rating, class ranking of student(Excellent, Average, Bad).

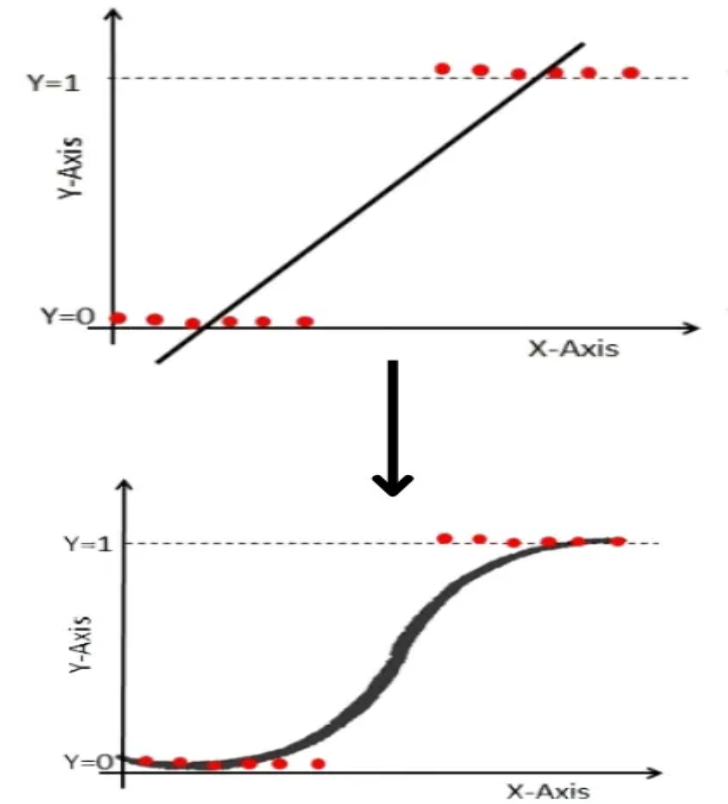
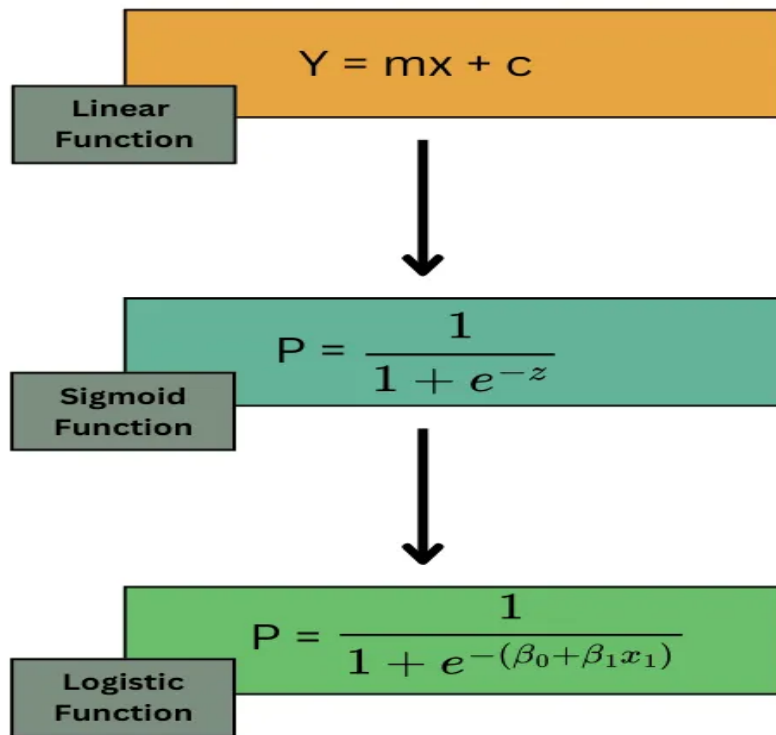
Setting up the logistic regression equation

- In logistic regression the target value is either 0 or 1 and predicted value ($\hat{y}$) is the probability varies within $[0, 1]$, but if we use linear regression this probability may exceed 1 or go below 0 varies within $(-\infty, \infty)$.
- How to map: $P \leftarrow \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$
- Let us consider $z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$
- $\frac{P}{1-P} \leftarrow z$; odds $\frac{P}{1-P}$ varies from $[0, \infty)$
- $\log_e \left(\frac{P}{1-P} \right) \leftarrow z$; log-odds (also called the logit function) of $y=1$ varies as a linear function of the input features within $(-\infty, \infty)$.
- Now we can simplify log-odds to $\frac{P}{1-P} \leftarrow e^z$

Sigmoid function: Transform the log-odds back to a probability

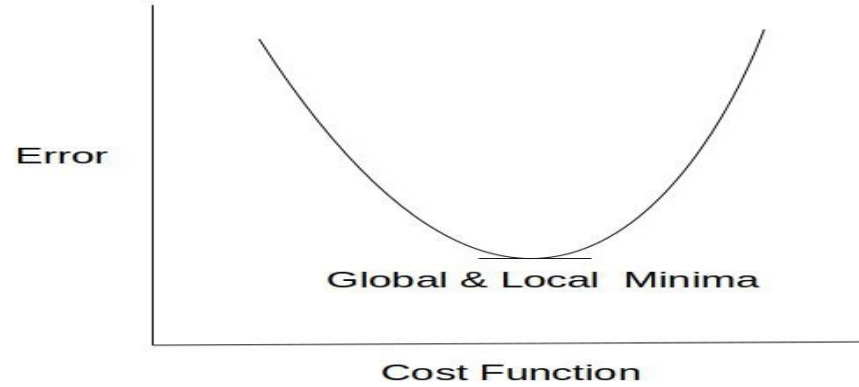
- $P = \frac{e^z}{1 + e^z}$
- $P(1 + e^z) = e^z \Rightarrow P = \frac{e^z}{1 + e^z} \Rightarrow P = \frac{1}{1 + e^{-z}}$
- $\sigma(z) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p)}}$
- where $\sigma(z)$ is the Sigmoid or logistic function
- $\hat{y} = P(y = 1|x) = \sigma(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p)$
- Thus, the probability of $y = 0$ is: $P(y = 0|x) = 1 - P(y = 1|x)$

Sigmoid Function



Cost Function in Logistic Regression

- In linear regression, we use the Mean squared error as the cost function which can be shown as:



- In logistic regression $\hat{y}_i$ is a non-linear function ($\hat{y} = 1 / (1 + e^{-z})$). If we use this in the above MSE equation, then it will give a non-convex graph with many local minima as shown



Likelihood function

- When we convert linear regression as binary classifier, it is known as Logistic regression.
- For binary classification, the output y (true value) is binary: $y \in \{0, 1\}$
- Logistic regression predicts the probability of categorical dependent variable (y) as probability of an event (Class 1) as follows:

$$\hat{y} = P(y = 1|x)$$

- The probability of $y = 0$ is as follows:

$$\hat{y} = P(y = 0|x) = 1 - P(y = 1|x)$$

- Given n independent observations $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$, the likelihood function is: $\mathcal{L}(\beta, \beta_0) = \prod_{i=1}^N P(y_i|x_i)$

Log-Likelihood or Cross-Entropy Loss Function

- A different cost function is used for logistic regression called Log-likelihood function which is derived from Likelihood function.
- To estimate the coefficients (β_i) of logistic regression, the maximum likelihood estimation method is used.
- The likelihood function is expressed for all independent observations N as
- $\mathcal{L}(\beta, \beta_0) = \prod_{i=1}^N P(y_i|x_i) \rightarrow \prod_{i=1}^N \hat{y}_i^{y_i} \cdot \hat{y}_i^{1-y_i}$
- Taking the logarithm of the likelihood to obtain the log-likelihood:
- $\log_e \mathcal{L}(\beta, \beta_0) = \sum_{i=1}^N [y_i \log_e \hat{y}_i + (1 - y_i) \log_e (1 - \hat{y}_i)]$

Negative Log-Likelihood Function

- Log-likelihood cost function is a concave function but negative of log-likelihood, which is the cost function for logistic regression, is the negative of a concave function.
- The negative of a concave function is a convex function.
- Final cost function $J(\beta)$ of logistic regression is negative of log-likelihood function as follows:
- $$J(\beta) = -\frac{1}{N} \sum_{i=1}^N [y_i \log_e \hat{y}_i + (1 - y_i) \log_e (1 - \hat{y}_i)]$$
- Minimizing the negative log-likelihood is a most common optimization problem.
- Gradient descent algorithm is designed to minimize cost function $J(\beta)$ rather than maximize log-likelihood function.
- This ensures that gradient descent algorithm will converge to the global minimum of the cost function.

Compute the Gradient of the Cost Function

- To minimize the cost function, we compute its gradient with respect to the weights w . The derivative of the cost function of a single data point:

- $$\frac{\partial J}{\partial \beta} = \frac{\partial J}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \times \frac{\partial z}{\partial \beta}$$

$$\frac{\partial J}{\partial \hat{y}} = -\frac{y}{\hat{y}} + \frac{1-\hat{y}}{1-\hat{y}} = \frac{-y + \hat{y} + \hat{y} - \hat{y}^2}{\hat{y}(1-\hat{y})} = \frac{\hat{y} - y}{\hat{y}(1-\hat{y})}$$

$$\begin{aligned} \frac{\partial \hat{y}}{\partial z} &= +1 \cdot \frac{1 * e^{-z}}{(1+e^{-z})^2} = \frac{e^{-z}}{(1+e^{-z})^2} \cdot \frac{1}{(1+e^{-z})} \\ &= \frac{1+e^{-z}-1}{(1+e^{-z})} * \hat{y} \\ &= \hat{y} * \left(\frac{1+e^{-z}}{1+e^{-z}} - \frac{1}{1+e^{-z}} \right) \\ &= \hat{y} * (1 - \hat{y}) \end{aligned}$$

$$\frac{\partial z}{\partial \omega} = x$$

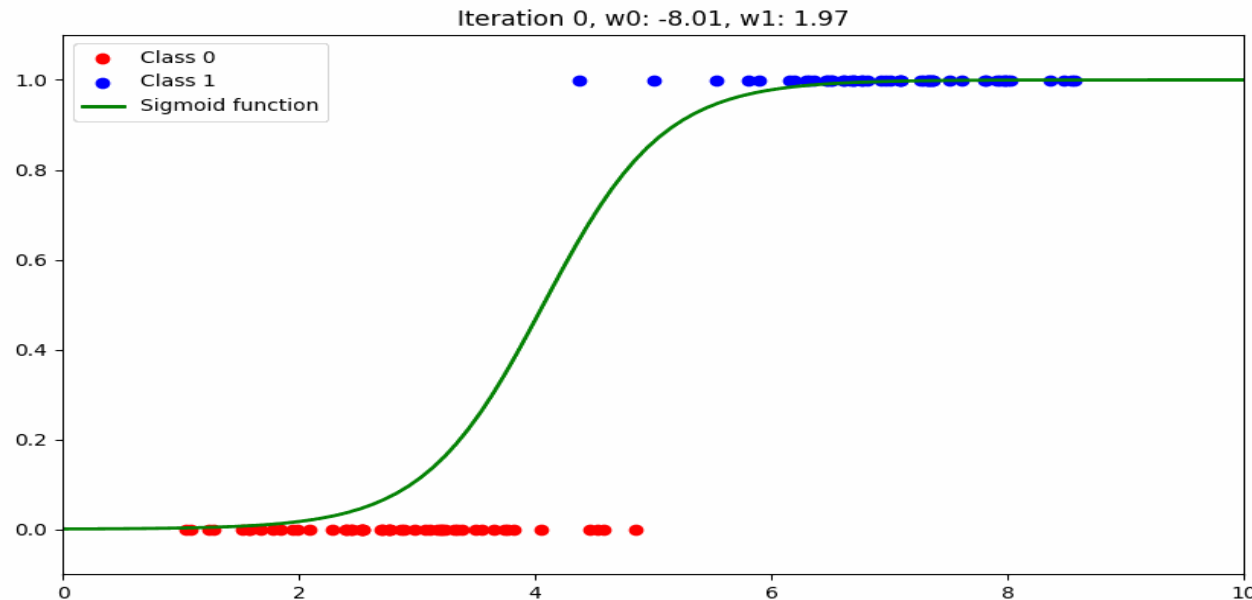
$$\frac{\partial J}{\partial \beta} = \frac{\hat{y} - y}{\hat{y}(1 - \hat{y})} \times \hat{y}(1 - \hat{y}) \times x = (\hat{y} - y)x$$

Update Coefficients using Gradient Descent

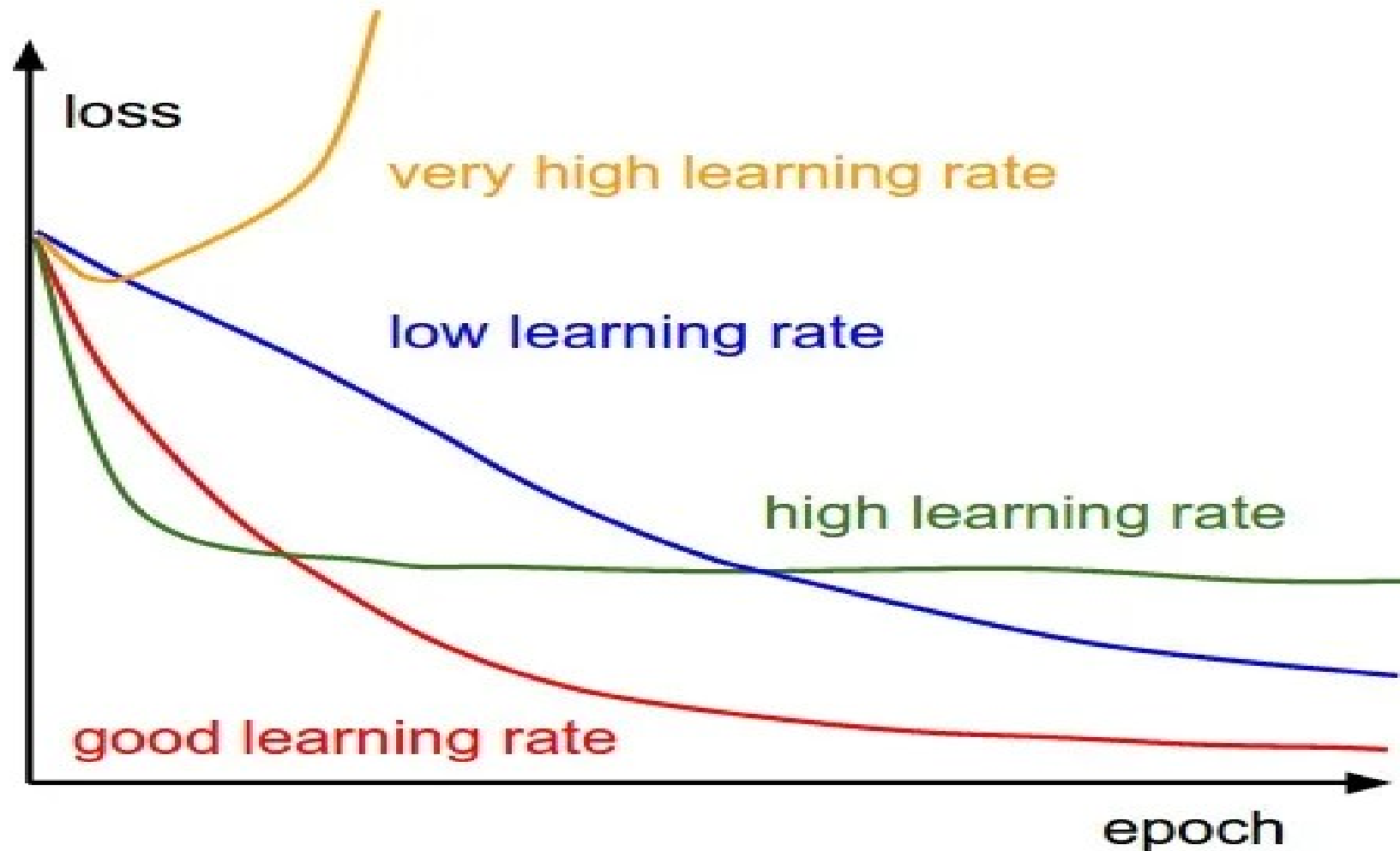
- After the derivatives are calculated, Using gradient descent, we update the coefficients of logistic regression as follows equation:
- $\beta_{new} = \beta_{old} - \eta \frac{\partial J}{\partial \beta}$
- $\beta_{new} = \beta_{old} - \eta(\hat{y} - y)\mathbf{x}$
- $\beta_{new} = \beta_{old} + \eta \frac{1}{N} X^T (\hat{y} - y)$
- Here η is the learning rate that controls the updates from being too large (which could cause the algorithm to overshoot the minimum) or too small (which could make convergence very slow).
- Therefore, finding the optimal learning rate is crucial, and this is typically selected through experimentation.

Coefficients (β) update stiffness of the Sigmoid

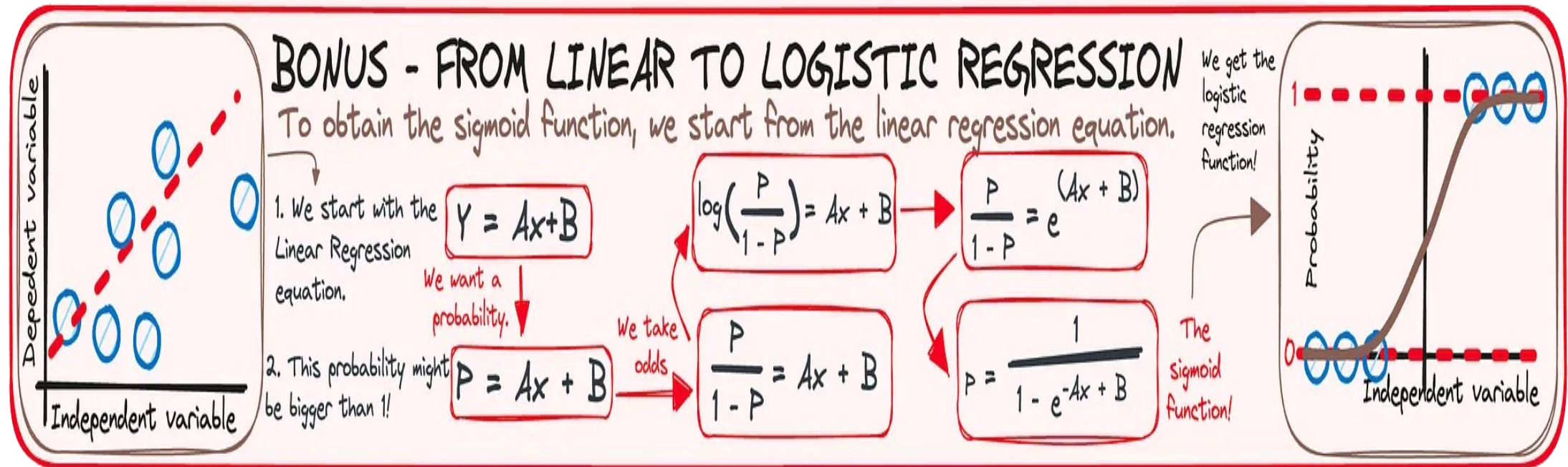
- The stiffness of the sigmoid function in logistic regression is directly influenced by the magnitude of the coefficients.
- Larger coefficients make the sigmoid stiffer (steeper), resulting in sharper decision boundaries and higher confidence in predictions.
- Smaller coefficients lead to a flatter sigmoid, resulting in smoother decision boundaries and lower confidence.



Effect of Learning Rate (η)



From Linear Regression to Logistic Regression



Questions

Hour (x)	Pass/Fail (y)
29	0
15	0
33	1
28	1
39	1

$$\log(\text{odds}) = z = -64 + 2 * \text{hours}$$

Q1. Calculate the probability of pass for the student who studied 33 hours: **Ans 88%**

Q2. At least how many hours a student should study that makes he will pass the course with the probability of more than 95%: **Ans: 33.47 hours**

Questions.....

- Some samples of two classes of articles: Technical (1) and Non-technical (0) are given.
- Each class has two features:
 - Time, which represent the average time required to read an article in hours,
 - Sentences, representing a number of sentences in a book
- first, we need to train our logistic regression model.

Time (Hr)	Sentences (1= 1000)	Article Type
2.7	2.5	0
1.4	2.3	0
3.3	2.4	0
1.3	1.8	0
3	3	0
7.6	2.7	1
5.9	2.2	1
6.9	1.8	1
8.6	3.5	1
7.7	3.5	1

1.9

3.1

?

Questions.....

- After 20 iteration, we get:.....

$$B0 = -0.1068913$$

$$B1 = 0.41444855$$

$$B2 = -0.2486209$$

- Thus, the decision boundary is given as:

Z = <multiple linear regression line>

Time (Hr)	Sentences (1= 1000)	Article Type
2.7	2.5	0
1.4	2.3	0
3.3	2.4	0
1.3	1.8	0
3	3	0
7.6	2.7	1
5.9	2.2	1
6.9	1.8	1
8.6	3.5	1
7.7	3.5	1

1.9

3.1

?

Questions.....

- After 20 iteration, we get:.....

$$B0 = -0.1068913$$

$$B1 = 0.41444855$$

$$B2 = -0.2486209$$

- Thus, the decision boundary is given as:

$$Z = B0 + B1 * X1 + B2 * X2$$

$$Z = -0.1068913 + 0.41444855 * \text{Time} - 0.2486209 * \text{Sentences}$$

Time (Hr)	Sentences (1= 1000)	Article Type
2.7	2.5	0
1.4	2.3	0
3.3	2.4	0
1.3	1.8	0
3	3	0
7.6	2.7	1
5.9	2.2	1
6.9	1.8	1
8.6	3.5	1
7.7	3.5	1

1.9

3.1

?

Questions

- For, $X_1 = 1.9$ and $X_2 = 3.1$, we get: . .

$$Z = -0.101818 + 0.41444855 * 1.9 - 0.2486209 * 3.1$$

$$Z = -0.085090545$$

- Now, we use sigmoid function to find the probability and thus predicting the class of given variables.

$$y = \frac{1}{1 + e^{-Z}} \quad \text{Or, } y = \frac{1}{1 + e^{-(-0.085090545)}}$$

- As $y = 0.477$, is less than 0.5, we can safely classify given sample to class Non-technical.

Time (Hr)	Sentences (1= 1000)	Article Type
2.7	2.5	0
1.4	2.3	0
3.3	2.4	0
1.3	1.8	0
3	3	0
7.6	2.7	1
5.9	2.2	1
6.9	1.8	1
8.6	3.5	1
7.7	3.5	1

1.9

3.1

?

Questions.....

Let's start off by assigning 0.0 to each coefficient and calculating the probability of the first training instance that belongs to class 0.

$$B_0 = 0.0$$

$$B_1 = 0.0$$

$$B_2 = 0.0$$

The first training instance is: $x_1=2.7810836$, $x_2=2.550537003$, $Y=0$

Calculate the coefficients after 1 epochs of stochastic gradient descent.